QUESTION PAPER AND SOLUTIONS

CST206 Operating Systems - End-Semester Previous Examination Paper with Solutions

Complete Question Bank, Scheme of Valuation & Worked Solutions (2025/2026)

Prescribed Institution / Board:	APJ Abdul Kalam Technological University (KTU)
Degree / Academic Level:	B.Tech in Computer Science and Engineering (Semester 4)
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Author / Academic Department:	Office of the Controller of Examinations, KTU
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Part A: Mandatory Short Answer Questions (3 Marks Each)

Question 1: Process State Transition Diagram

Explain the 5-state process lifecycle model with an annotated transition graph.

Solution: A process transitions through NEW, READY, RUNNING, WAITING (BLOCKED), and TERMINATED states. The CPU scheduler dispatches ready processes to running. Interrupts transition running to ready. I/O requests move processes to waiting.

Question 2: Critical Section Problem & Petersons Solution

Define the three essential requirements for any valid solution to the critical section problem.

Solution: 1. Mutual Exclusion: If process Pi is executing in its critical section, no other processes can be executing. 2. Progress: If no process is executing and some wish to enter, selection cannot be postponed indefinitely. 3. Bounded Waiting: A bound must exist on the number of times other processes enter after a request is registered.

Question 3: Beladys Anomaly in Page Replacement

Explain Beladys Anomaly with an illustrative example.

Solution: Beladys Anomaly refers to the counter-intuitive phenomenon where allocating more physical page frames results in an increased number of page faults. It occurs in FIFO replacement but is mathematically impossible in stack-based algorithms like LRU and Optimal.

Part B: Analytical & Numerical Questions (14 Marks Each)

Question 4: CPU Scheduling Algorithms Comparative Problem

Given five processes P1(burst 10, arrival 0), P2(burst 1, arrival 1), P3(burst 2, arrival 2), P4(burst 1, arrival 3), P5(burst 5, arrival 4): Compute Average Turnaround Time and Waiting Time under Preemptive SJF (SRTF).

Solution Steps: Gantt chart executes P1 [0-1], P2 [1-2], P4 [2-3], P3 [3-5], P5 [5-10], P1 [10-19].

Completion Times: P1=19, P2=2, P3=5, P4=3, P5=10. Average Turnaround Time = (19 + 1 + 3 + 0 + 6) / 5 = 5.8 ms. Average Waiting Time = (9 + 0 + 1 + 0 + 1) / 5 = 2.2 ms.

Question 5: Bankers Algorithm for Deadlock Avoidance

Given 5 processes and 3 resource types (A=10, B=5, C=7), demonstrate if the system is in a safe state and output the complete Safe Sequence.

Solution: Calculating Need Matrix = Max - Allocation. Safe Sequence generated: < P1, P3, P4, P0, P2 >. The state is verified safe; no deadlock can occur.